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Improving the surface charge density of a contact-separation-based triboelectric nanogenerator by modifying the surface morphology

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Abstract

A triboelectric nanogenerator (TENG) based on the contact-separation mode can be used as a portable power source to harvest the triboelectrification which arises between contacting films. However, a general challenge is that the charge generation is usually limited. The performance of a TENG can be enhanced by modifying the surface morphologies of the triboelectric materials. In this study, we demonstrate morphologies with line, pillar and hexagonal cone shapes on the surfaces of PDMS and PMMA to enhance the charge production in a TENG. A contact and subsequent separation operation between the two surfaces of triboelectric materials is then carried out to generate the charge. Experiments reveal that the maximum output power, voltage and current density of a TENG with 300-nm hexagonal cone patterns on the contacting surfaces are nearly 22, three and six times greater than a TENG without any patterns, respectively. In addition, the results showed that pillar patterns with a smaller width generate higher output voltage, current and power levels compared to those with larger widths. As a result, the generation of the surface charge density in a TENG is augmented by the creation of nanopatterns on its contacting elements. This result will be applicable to the fabrication of a self-powered nanosystem with high efficiency at a low cost.

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1. Introduction

The development of smart systems which collect mechanical energy from working environments and converting it into electricity has attracted increasing interest not only for meeting rapidly growing energy needs but also for achieving self-powered systems and to energize the portable electronics used in our everyday lives [1-4]. For these reasons, researchers have developed various approaches based on photovoltaics, electromagnetics, electrostatics, piezoelectrics, pyroelectrics, and other areas [5, 6]. Recently, the triboelectric nanogenerator (TENG) has emerged as an effective energy harvesting technology capable of utilizing random mechanical energy for powering nanodevices via triboelectrification and electrostatic induction [7, 8]. However, the increment of the charge generation in a TENG is a critical issue. In order to resolve this issue, the surface morphologies of the contacting materials are modified so that the charge production can increase by increasing the contact area and decreasing the effective thicknesses of the polydimethylsiloxane (PDMS) and polymethylmeth-acrylate (PMMA) used.

In early 2012, Fan et al. first introduced the concept of a TENG [9], after which the concept was expanded with regard to its potential applications, from energizing the micro/nano-electronic systems [10] to the realization of large-scale electricity production, such as 'blue' power. To achieve high-performance TENGs, several factors must be considered, including the contact area, contact force, contact speed, relative humidity and surface work function of the contacting materials [11]. Thus far, the most of the profound studies directed at improving the performance capabilities of TENGs have concentrated mainly on effective materials [12, 13] and device structures [14]. In contrast, less stress has been placed on the surface structure, although the structural features of the contacting area can have a strong impact on the contact-separation-based charge generation and triboelectrification characteristics. Therefore, there is a strong need for intense research on the effects of nanoscale surface morphologies which can significantly influence the contacting features and device outcomes [15].

This study seeks to improve the surface charge density of a TENG by modifying the surface morphology of its contacting materials experimentally. Consequently, the problem of limited surface charge generation due to the natural phenomenon associated with the triboelectric effects of flat surfaces can be overcome. PDMS and PMMA triboelectric materials are chosen owing to their excellent chemical strengths,

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moderate relative permittivity levels, low weights, and simple handling conditions. In order to investigate the effects of the pattern sizes, pillar-shaped nanopattern arrays with four different pillar widths (200 nm, 250 nm, 300 nm and 350 nm) are fabricated using thermal nanoimprint lithography on the contacting areas of both triboelectric materials. An automated up-down movement system is used to realize periodic contact and separation between the PDMS and PMMA layers. The voltage, current, and power caused by the electrostatic charges resulting from the periodic contact and separation between the PDMS and PMMA are systematically investigated. Experiments demonstrated that the pillar-shaped nanopatterns with smaller widths generated higher output power, voltage, and current density levels compared to those with larger widths. In addition, line and hexagonal cone shaped surface morphologies with lengths of 300 nm are produced by thermal nanoimprint lithography to observe the charge formation and current flow on a TENG with different nanopattern shapes. It was found that the maximum output power, voltage and current density of a TENG with 300-nm hexagonal cone patterns on the contacting surfaces are almost 22, three and six times greater, respectively, than that of a TENG without any patterns. These results will be helpful for fabricating a self-powered nanosystem with high efficiency at a low cost.

2. Experiment

Thermal nanoimprint lithography (T-NIL) was used for replicating line, pillar and hexagonal cone nanopatterns on PMMA and PDMS substrates from a Si master stamp. The tests were performed under the conditions listed in Table 1. The results of the T-NIL process are shown in Figures 1 and 2. The values of the pattern widths and heights with different nanopattern shapes fabricated on the contacting areas of the TENG materials are shown in Table 2.

The theoretical model used in the experiment is shown in Figure 3. Based on the materials used in triboelectric nanogenerators, PDMS and PMMA were stacked in a face-to-face manner as two triboelectric layers. On the outer surface of these two materials, Al and Cu were attached as conducting electrodes. The contact and subsequent separation operation between the two surfaces of the triboelectric materials were carried out through the application of an automatic system with a rate of five cycles per second. The contact force introduced between the PDMS and the PMMA was monitored and regulated by a force sensor. After coming into contact, the inner surfaces of the layers induced opposite static charges (σ) with a uniform

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distribution. The layers were then separated from each other by an increased value of x(t) and a potential difference (V) between Al and Cu was created. Subsequently, electrical Cu wire served to connect the conductors such that a current flow was created based on a surface-state model [16].

A single cycle of the operation of the contact-mode TENG can be explained by the coupling of contact charging and an electrostatic effect [17], as sketched in Figure 4. At the initial position, no charge is generated as the contacting materials are fully separated. The back and forth motor movement brings the PDMS and PMMA into contact, creating positive triboelectric charges on the PMMA and negative charges on the PDMS side (Figure 4 (a)) due to triboelectrification. According to the triboelectric series [18], a material toward the bottom of the series, when making contact with a material near the top of the series, will attain a more negative charge. According to this rule, electrons which are injected from the PMMA into the PDMS are retained on the PDMS; thus, the PMMA becomes an electrically positively charged material and the PDMS becomes a negatively charged material. Afterward, during the separating condition of the TENG (Figure 4 (b)), an electron moves from the Cu film to the Al film due to the generation of a potential difference between the conductors. If ε_0 is the vacuum permittivity, S is the contact area, σ is the generated triboelectric charge density, and $v(t)$ is the contact speed at a given state, the open-circuit voltage and the short-circuit current generated across the output terminal can then be defined by [19]

$$V_{oc} = \frac{\sigma x(t)}{\varepsilon_0}, \qquad (1)$$

$$I_{sc} = \frac{S\sigma d_0 v(t)}{(d_0 + x(t))^2}, \qquad (2)$$

where, d_0 is calculated by

$$d_0 = \frac{\varepsilon_{r2} d_1 + \varepsilon_{r1} d_2}{\varepsilon_{r1}\varepsilon_{r2}}. \qquad (3)$$

In this equation, ε_{r1} and ε_{r2} are the relative permittivity of PDMS and PMMA, and d_1 and d_2 represent the thickness of the PDMS and the PMMA, respectively. The maximum current flow is obtained when the two polymers are fully separated (Figure 4 (c)). In the contact condition of the TENG (Figure 4 (d)), the electron flow occurs in a reversed direction due to the generation of electric potential with reversed polarity. It should be noted that the insulating PDMS allows the long-term retention of triboelectric charges on its surfaces [17] despite the fact that the triboelectric charges on the metal sides are periodically screened by inductive charges. During this process, the TENG acts as an electron pump that drives electrons back and forth

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between the two electrodes [18].

3. Results and Discussion

3.1. Output characterization of a TENG with various nanopattern shapes

To investigate the performance of the TENG, three sets of experiments were conducted to evaluate the charge transfer behavior between the PMMA and the PDMS. At fixed triggering speeds and under fixed amounts of force in the system, electrical output measurements were conducted under the following three conditions: a) without a pattern on the material surfaces, b) with a 150-nm line pattern on the material surfaces, and c) with a 300-nm hexagonal cone pattern on the material surfaces as shown in Figure 5. The effective contact area of the TENG is 4×4 cm^2, which produces an open-circuit voltage (V_{oc}) of 129.76 V without a pattern (Figure 5 (a)), of 263.03 V with a 150-nm line pattern (Figure 5 (b)), and of 451.75 V with a 300-nm hexagonal cone pattern (Figure 5 (c)) on both the PDMS and the PMMA surfaces. The voltage output is nearly doubled in the hexagonal cone patterns as compared to the line patterns, with a threefold increase over the TENG without a pattern. Given that electrons cannot flow back to mitigate the induced electric potential difference between the two electrodes under the open-circuit condition, the voltage remains at a plateau until the next contact emerges [10-12]. Meanwhile, the peak values of the short-circuit current density reach 37.51 $\mu A/m^2$ without a pattern (Figure 6 (a)), 121.81 $\mu A/m^2$ with the line pattern (Figure 6 (b)), and 237.28 $\mu A/m^2$ with the hexagonal cone pattern (Figure 6 (c)) on both polymer faces. The faster approach is expected to produce larger current peaks than the slower separation process.

In addition, to investigate the charge transfer behavior between the PMMA and PDMS in the TENG, four sets of experiments were conducted at fixed triggering speeds and fixed forces of the automatic system (five cycles per second). The maximum power levels of the TENG with no pattern and with the pillar pattern, line pattern and hexagonal cone pattern on both materials were found to be 4.87 mW/m^2, 17.86 mW/m^2, 31.67 mW/m^2 and 107.19 mW/m^2, respectively, as shown in Figure 7. It was also noted that the power output in the TENG with 150-nm lines is 6.5 times greater than that in the TENG without a pattern, while the power output in the TENG with 300-nm hexagonal cone pattern is 22 times higher than that in the TENG without a pattern.

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3.2. Output characterization of the TENG with various nano-sized pillar widths

Four different (200 nm, 250 nm, 300 nm and 350 nm) widths of the pillar-shaped nanopatterned TENGs were fabricated to compare their performances. The measured open-circuit voltages (V_{oc}) were found as shown in Figure 8 to be 199.44 V, 175.53 V, 158.47 V, and 134.91 V in the pillar-shaped TENGs with 200nm, 250 nm, 300 nm and 350 nm widths, respectively. This indicates that smaller widths with pillar-shaped nanopatterns generate higher output voltage levels compared to those generated by larger widths in pillar-shaped nanopatterned TENGs.

Figure 9 presents the short-circuit current density levels of TENGs with various pillar widths. The highest value of the current density was achieved with the 200-nm pillars widths; at 139.68 $\mu A/m^2$, whereas the smallest value of the current was obtained from the 350-nm pillar widths, with a value of 58.31 $\mu A/m^2$. Accordingly, pillar-shaped nanopatterns with smaller widths generated higher output current density levels compared to those with larger widths.

Figure 10 illustrates the results of a comparison of the power output levels in TENGs with various pillar widths. It was determined that the power output of the TENG patterned with the 200 nm pillars (27.86 mW/m^2) was nearly six times higher than that of unstructured film samples (4.87 mW/m^2). In contrast, the maximum power levels for the TENGs with pillar patterns of 250 nm, 300nm and 350nm were found to be 21.74 mW/m^2, 12.85 mW/m^2 and 7.92 mW/m^2, respectively.

3.3. Dependence of the TENG power under an external load

Resistors were utilized as external loads to investigate the output instantaneous peak power of the TENGs further with all three nanopattern shapes. As shown in Figure 11, the overall instantaneous peak power (I^2_{peak} R) levels for the cases without a pattern, with the line, pillar and hexagonal cone patterns were found to be maximized at a load resistance of 7 MΩ. Moreover, the peak power increases dramatically in the TENG with the hexagonal cone pattern, reaching 0.117 W.

4. Conclusion

Triboelectric energy harvesters are developed based on nanoscale surface morphologies fabricated by the nanopatterning technique known as thermal nanoimprint lithography. Surface structural variations on the nano-scale range provided control over the levels of the contact area and friction, factors which are crucial for efficient triboelectrification. The direct electrical

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relationships among TENGs without a pattern and with a pillar pattern, a line pattern and a hexagonal cone pattern on dielectric films were investigated in experiments. Considering various pattern shapes, the experiments revealed that the maximum output power of a TENG with 300-nm hexagonal cone patterns on the contacting surfaces is nearly 22 times greater than that by a TENG without any patterns. It was also found that the output voltage generated in the TENG with the hexagonal cone pattern is nearly double that generated with the line pattern and is three times that with a flat surface. In addition, the peak current density in the nanogenerator with the hexagonal cone features achieved a value of 237.28 $\mu A/m^2$, which is six that obtained by the nanogenerator without a pattern.

Considering various pillar widths, It was determined that the power output of the TENG patterned with the 200-nm pillars (27.86 mW/m^2) is nearly six times higher compared to that by unstructured films. It was also found that the maximum output current and voltage of the TENG with the 200-nm pillar morphology on the contacting areas are nearly four times and 1.5 times greater, respectively than those of a TENG without any patterns. Finally, the instantaneous power in each TENG was observed to be maximized at a load resistance of 7 MΩ. These experimental results will serve as a stepping stone for the development of high-efficiency and cost-effective self-powered nano-devices in the near future.

Acknowledgement

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References

[1] F. R. Fan, L. Lin, G. Zhu, W. Z. Wu, R. Zhang and Z. L. Wang, “Transparent triboelectric nanogenerators and self-powered pressure sensors based on micropatterned plastic films”, Nano Lett., 12 (**2012**) 3109–3114

[2] Z. L. Wang, “Triboelectric nanogenerators as new energy technology for self-powered systems and as active mechanical and chemical sensors”, ACS Nano, 7 (**2013**) 9533−9557.

[3] G. Zhu, C. F. Pan, W. X. Guo, C. Y. Chen, Y. S. Zhou, R. M. Yu and Z. L. Wang, “Triboelectric-generator-driven pulse electrodeposition for micropatterning”, Nano Lett. 12 (**2012**) 4960–4965.

[4] S. Wang, L. Lin and Z. L. Wang, “Nanoscale

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triboelectric-effect enabled energy conversion for sustainably powering portable electronics", Nano Lett. 12 (**2012**) 6339–6346.

[5] R. G. Horn, D. T. Smith and A. Grabbe, "Contact electrification induced by monolayer modification of a surface and relation to acid–base interactions", Nature 366 (**1993**) 442–443.

[6] R. G. Horn and D. T. Smith, "Contact electrification and adhesion between dissimilar materials", Science 256 (**1992**) 362–364.

[7] L. S. McCarty and G. M. Whitesides, "Electrostatic charging due to the separation of ions at interfaces: contact electrification of ionic electrets", Angewandte Chemie 47 (**2008**) 2188–2207.

[8] H. T. Baytekin, A. Z. Patashinski, M. Branicki, B. Baytekin, S. Soh and B. A. Grzybowski, "The mosaic of surface charge in contact electrification", Science 333 (**2011**) 308–312.

[9] F. R. Fan, Z. Q. Tian and Z. L. Wang, "Flexible Triboelectric Generator!", Nano Energy 1 (**2012**) 328–334.

[10] G. Zhu, Z. H. Lin, Q. Jing, P. Bai, C. Pan, Y. Yang, Y. Zhou and Z. L.Wang, "Toward large-scale energy harvesting by a nanoparticle-enhanced triboelectric nanogenerator", Nano Lett. 13 (**2013**) 847–853.

[11] P. Bai, G. Zhu, Z. H. Lin, Q. S. Jing, J. Shen, G. Zhang, J. S. Ma and Z. L. Wang, "Integrated multilayered triboelectric nanogenerator for harvesting biomechanical energy from human motions", ACS Nano 7 (**2013**), 3713–3719.

[12] S. Kim, M. K. Gupta, K. Y. Lee, A. Sohn, T. Y. Kim, K. S. Shin, D. Kim, S. K. Kim, K. H. Lee, H. J. Shin, D. W. Kim and S. W. Kim, "Vertically stacked thin triboelectric nanogenerator for wind energy harvesting", Adv. Mater., 26 (**2014**) 3918−3925.

[13] Z. H. Lin, G. Cheng, L. Lin, S. Lee and Z. L. Wang, "Dual-mode triboelectric nanogenerator for harvesting water energy and as a self-powered ethanol nanosensor", Angew. Chem., 52 (**2013**) 12545−12549.

[14] M. Taghavi, V. Mattoli, A. Sadeghi, B. Mazzolai and L. Beccai, "A novel soft metal-polymer composite for multidirectional pressure energy harvesting", Adv. Energy Mater, 4 (**2014**) 1400024.

[15] X. S. Zhang, M. D. Han, R. X. Wang, F. Y. Zhu, Z. H. Li, W. Wang and H. X. Zhang, "Frequency-multiplication high output triboelectric nanogenerator for sustainably powering biomedical Microsystems", Nano Lett. 13 (**2013**) 1168–1172.

ACCEPTED MANUSCRIPT

[16] A.W. Cowley and S.M. Sze, “Surface states and barrier height of metal-semiconductor systems”, Journal of Physics, 36 (**1965**) 3212.

[17] W.R. Harper, “The generation of static charge”, Advances in Physics, 6(24) (**1957**) 365-417.

[18] A. F. Diaz and R. M. Felix-Navarro, “A semi-quantitative triboelectric series for polymeric materials: the influence of chemical structure and properties”, J. Electrostat. 62 (**2004**), 277–290.

[19] Niu, S., Wang, S., Lin, L., Liu Y., Zhou, Y., Hu, Y., and Wang, Z., “Theoretical study of contact-mode triboelectric nanogenerators as an effective power source”, Energy & Environmental Science, 6 (**2013**), 3576-3583.

Figures List

Figure 1 SEM images of TENG materials with different pattern shapes. (a) Line pattern, (b) Pillar pattern, and (c) Hexagonal cone pattern.

Figure 2 SEM images of TENG materials with different widths. (a) 200 nm, (b) 250 nm, (c) 300 nm, and (d) 350 nm.

Figure 3 Theoretical model.

Figure 4 Schematic diagram of TENGs one operating cycle. (a) Fully contacted, (b) Separating, (c) Fully separated, and (d) Contacting.

Figure 5 Voltages of TENGs with different nano pattern shapes. (a) No patterns, (b) Line patterns, and (c) Hexagonal cone patterns.

Figure 6 Current densities of TENGs with different nano pattern shapes. (a) No patterns, (b) Line patterns, and (c) Hexagonal cone patterns.

Figure 7 Output power comparison of TENGs with different nano pattern shapes.

Figure 8 Open circuit voltages of TENGs with various pillar widths.

Figure 9 Short circuit current densities of TENGs with various pillar widths.

Figure 10 Output power comparison of TENGs with various pillar widths.

Figure 11 Peak powers of the TENGs at external loads.

Tables List

Table 1 Conditions for T-NIL.

Table 2 Pattern dimensions.

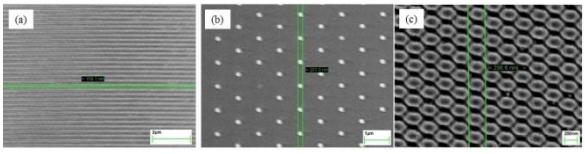


Figure 1

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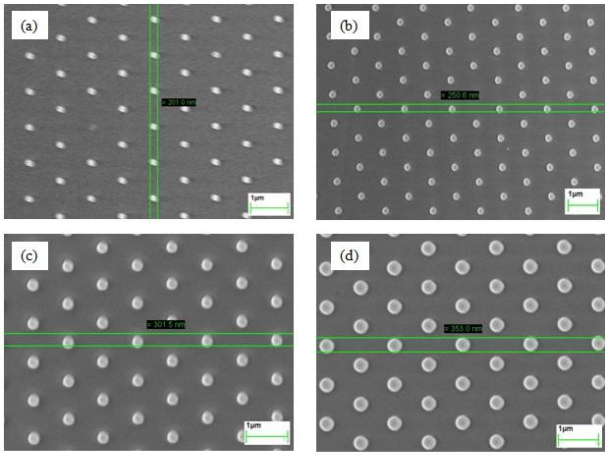


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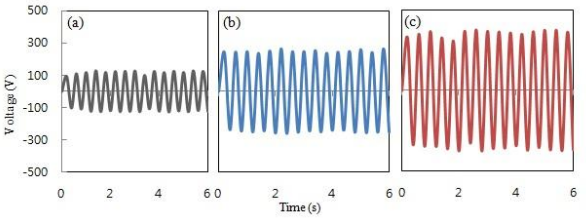


Figure 5

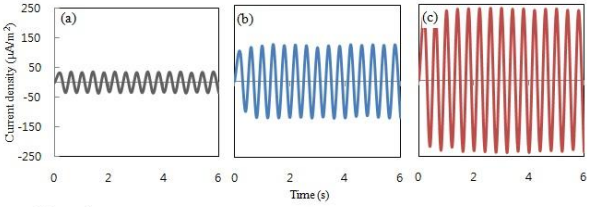


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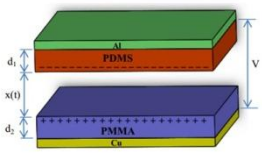


Figure 3

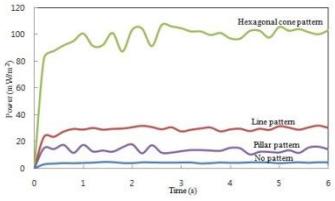


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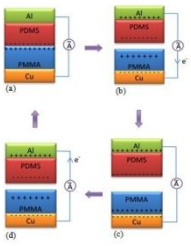


Figure 4

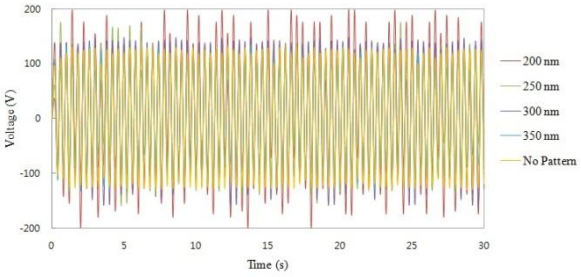


Figure 8

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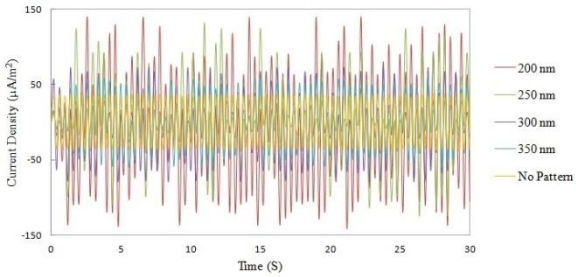


Figure 9

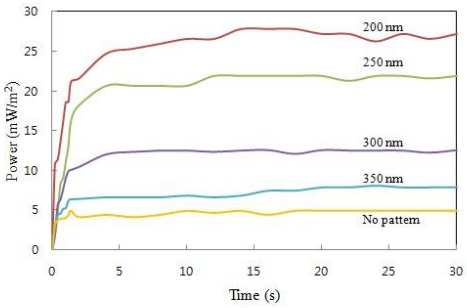


Figure 10

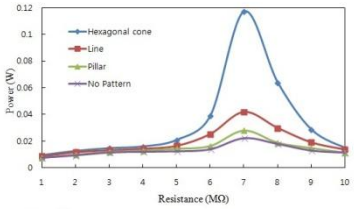


Figure 11

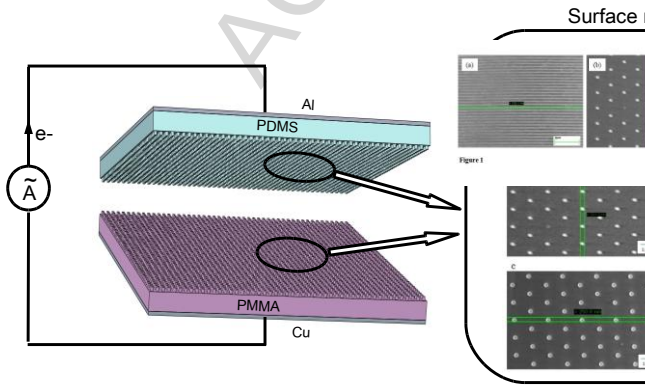
Table 1

No	Item	Condition
1	Spin-coating	3000 rpm for 30 s
2	Soft bake conditions	100°C for 5 min
3	Imprint temperature	150 °C for 10 min
4	Imprint load	20~30 kN
5	Curing time	10 min

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Table 2

	Pattern	Width or diameter (nm)	Height (nm)
1	No	-	-
2	Line	150	100
3	Pillar	200, 250, 300, 350	250
4	Hexagonal cone	300 (length) 200 (width)	150



Graphical abstract

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Highlights

- Effect of nano patterns on charge generation in TENG surfaces was investigated.
- Surface charge densities increased with the modification of contacting areas.
- Hexagonal cone shaped nano pattern produced higher output power than line and pillar pattern.
- Smaller width pillar patterns generated higher output voltage, current and power l evels compared to those with larger widths.
- Experimental results will aid for the development of high-efficiency and cost-effe ctive self-powered nano devices.

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